

A094 · Bridging Genome-Scale Metabolism and Adaptive Ecology: A Hybrid Consumer-Resource Framework for Dynamic Microbial Growth

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Microorganisms are shaped by both metabolic capabilities and ecological dynamics that emerge as organisms compete for, transform, and exchange resources. Most computational models capture only one scale, which is often inadequate to capture the full mechanisms of community dynamics. For instance, Consumer-Resource Models (CRMs) [1,2] describe ecological dynamics but lack biochemical resolution, whereas Genome-Scale Metabolic Models (GSMMs) [3-5] provide metabolic detail but often rely on static Flux Balance Analysis (FBA), limiting their use in fluctuating environments. Here, we introduce a hybrid genome-scale consumer-resource framework, CRM-FBA, that combines genome-scale metabolism with dynamic adaptive ecology. We first augment the microbial Consumer-Resource Model (MiCRM) from Marsland et al. [2] with adaptive uptake strategies from Pacciani-Mori et al. [1], to form an adaptive MiCRM. Because adaptive MiCRM lacks stoichiometric and genome-scale metabolic detail, we next coupled it to GSMMs through real-time FBA feedback. CRM-FBA reproduced diauxic shifts and predicted secretion and reconsumption of metabolic byproducts. We further scaled the CRM-FBA framework to microbial communities. In a 12-species synthetic gut consortium, we implemented a CRM-FBA community model in which genome-scale metabolism determined yields and cross-feeding stoichiometry while species-specific uptake kinetics were fit from data. We successfully predicted held-out leave-one-out community composition better than phenomenological CRM and a global-kinetics community GSMM model alone. We then validated this with a 10-species Db-MM consortium with measured community metabolites. We show that our framework successfully enabled us to predict microbial composition and function from this community. Together, CRM-FBA occupies the intersection that neither baseline CRM nor FBA reaches.